

Evaluating Patient Stratification for Heart Disease Classification: Clinical Subgroups, Data-Driven Clusters, and Global Modelling

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Contents

1	Introduction	2
2	Aims and Research Questions	3
2.1	Aim	3
2.2	Research Questions	3
2.3	Expected Outcomes	3
3	Methodology	4
3.1	Data and Variables	4
3.2	Data Preprocessing and Exploratory Analysis	4
3.3	Experimental Design	4
3.4	Model Development	5
3.5	Model Evaluation	5
4	Timeline	5

1 Introduction

Cardiovascular disease (CVD) is a leading cause of death worldwide, making accurate classification of patients with heart disease an important clinical and research problem.

Machine learning has been widely applied to heart disease classification using conventional, ensemble, and deep-learning classifiers [1], but further gains may not come from more complex classifiers alone. An alternative is to first group patients with similar characteristics and classify within each group. Data-driven clustering has been used in this way, for example through k-modes clustering [2] or comparisons of algorithms such as k-means and hierarchical clustering [3]. These studies, however, focus on particular clustering techniques rather than on whether data-driven grouping is more useful than clinically defined subgroups.

Patients can instead be stratified using established clinical characteristics before modelling. Sex-specific models have reported differences in predictive performance and important predictors between males and females [4, 5], and modelling a clinically defined population separately can address heterogeneity within a broader group [6]. Clinical stratification may therefore capture subgroup-specific relationships that a single population-wide model obscures.

Where the two approaches have been compared directly in other diseases, neither has been consistently superior: in type 2 diabetes, data-driven clusters identified distinct subgroups [7] yet simpler clinical features later proved more useful for stratification [8], while in acutely decompensated cirrhosis a data-driven framework distinguished groups that established clinical scores did not [9]. Their relative value must therefore be established empirically rather than assumed.

The two approaches involve different trade-offs: data-driven clustering can uncover groupings not tied to predefined clinical categories but depends on the variables and method used, whereas clinical stratification is more interpretable but may miss subtler patterns of similarity. In heart disease both have been applied but pursued separately, leaving it unclear whether data-driven clusters offer any advantage over clinically defined subgroups when both are evaluated against the same non-stratified baseline. This distinction matters because the way patients are grouped changes how heterogeneity is represented in a classification model, potentially affecting predictive performance and interpretability.

This study therefore investigates how clinically defined subgroups and data-driven clusters affect heart disease classification. A common non-stratified model provides a global baseline, against which classification models built on clinically defined subgroups and on data-driven clusters are compared under

matched conditions – the same features, classifier family, preprocessing, and evaluation metrics. This determines whether stratification improves classification and whether the way patient groups are defined affects performance.

2 Aims and Research Questions

2.1 Aim

The aim of this project is to evaluate how patient stratification affects heart disease classification by comparing a global, non-stratified model, clinically defined subgroups, and data-driven clusters under a common experimental framework. Because the three approaches use the same dataset, classifiers, preprocessing, and evaluation protocol, observed differences can be attributed primarily to the stratification strategy.

2.2 Research Questions

- RQ1.** How does stratifying patients into clinically defined subgroups affect heart disease classification performance, compared with a global, non-stratified model?
- RQ2.** How does stratifying patients into data-driven clusters affect heart disease classification performance, compared with a global, non-stratified model?
- RQ3.** How do clinically defined subgroups and data-driven clusters differ in their effect on heart disease classification performance?

2.3 Expected Outcomes

The study will provide a controlled comparison of the three approaches under the same features, classifiers, preprocessing, and evaluation procedure. The outcome will not be assumed in advance: evidence from other diseases shows that data-driven subgroups do not necessarily outperform clinical stratification [8], so any of the three approaches may perform best, or differences between them may be limited.

The intended contribution is therefore a controlled comparison of patient-stratification strategies for heart disease classification, rather than the development of another classification model. Whether this specific comparison has been conducted previously will be confirmed through the literature review rather than assumed.

3 Methodology

3.1 Data and Variables

The study will use the Cleveland subset of the UCI Heart Disease dataset [10], a widely used benchmark that enables comparison with previous work [1]. It contains 303 patient records with 13 clinical features, including age, sex, chest pain type, blood pressure, cholesterol, electrocardiographic results, maximum heart rate, and exercise-induced angina. The target is a binary label representing the presence or absence of heart disease.

3.2 Data Preprocessing and Exploratory Analysis

Missing values will be handled using a documented procedure, duplicate records removed, categorical variables encoded, and continuous features standardised where required; outliers will be retained unless clearly erroneous, given the small sample. To prevent data leakage, all preprocessing steps that estimate parameters will be fitted on training data only. Exploratory analysis will cover summary statistics, class distribution, feature and missing-value patterns, and the sizes of the candidate subgroups and clusters to ensure that each group is sufficiently represented for reliable training and evaluation.

3.3 Experimental Design

The study compares three approaches that share the same features, classifiers, preprocessing procedure, and evaluation protocol, differing only in how patient heterogeneity is handled.

Arm A – Global model (baseline). A single classifier is trained on the full patient population without stratification.

Arm B – Clinically defined subgroups. Patients are partitioned into clinically defined subgroups, and a classifier is trained within each. Sex is selected as the stratification variable because cardiovascular disease can differ between males and females in risk-factor profiles and feature–outcome relationships, with sex-specific models reporting differences in predictive performance and important predictors [4, 5]. As a binary variable, sex also produces only two subgroups, helping maintain sufficient observations for model training given the limited sample size.

Arm C – Data-driven clustering. Patients are grouped by an unsupervised algorithm, and a classifier is trained within each cluster. Because the dataset contains both continuous and categorical clinical variables, k-prototypes is used because it can accommodate both feature types, whereas

k-means and k-modes are designed primarily for continuous and categorical data respectively. The use of k-prototypes is therefore aligned with the structure of the dataset and the objective of identifying patient groups from mixed clinical features. The number of clusters is kept small and selected using an internal criterion such as silhouette analysis, with fixed random seeds for reproducibility.

Given the approximately 303 records, the number of subgroups and clusters is deliberately limited to avoid groups that are too small for reliable training. Group sizes will be checked during exploratory analysis, and the cluster count reduced if a resulting partition contains insufficient observations.

Arm B versus Arm A addresses RQ1, Arm C versus Arm A addresses RQ2, and Arm B versus Arm C addresses RQ3.

3.4 Model Development

Two classifiers are used across all arms: logistic regression as an interpretable linear model and random forest as a non-linear comparator [1]. The same classifiers and hyperparameter-selection procedure are applied in every arm, so that performance differences reflect the stratification strategy rather than the choice of classifier. Consistent with the study’s aim, no further classifiers or extensive optimisation are pursued.

3.5 Model Evaluation

All arms are evaluated with stratified k -fold cross-validation using the same fold partitions. Performance is reported using accuracy, F1-score, and ROC-AUC. Accuracy supports comparison with previous heart disease studies, while F1-score and ROC-AUC provide additional measures of classification performance under potentially uneven class distributions.

For the data-driven arm, clustering will be fitted using the training portion of each cross-validation fold so that validation data do not influence cluster formation. Predictions from the subgroup and cluster models are pooled to give population-level estimates directly comparable with the global model. Mean performance and variability across folds are reported for each arm, allowing differences between strategies to be assessed rather than relying on single point estimates.

4 Timeline

Task	1	2	3	4	5	6	7	8	9	10	11	12
Initial meeting with supervisor		•										
Literature review & writing proposal		•	•									
Data preparation & preprocessing			•	•								
Global model & evaluation (Arm A)				•	•							
Clinically defined subgroups & evaluation (Arm B)					•	•						
Data-driven clustering & evaluation (Arm C)						•	•					
Comparison across arms							•	•	•			
Analysis & interpretation									•	•		
Report writing								•	•	•	•	
Revision & final submission											•	•

Table 1: Project timeline

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